

Histology, Part Two. Fibrosis and Stains.

In the last chapter we learned to read the pattern of injury on a liver biopsy. In this chapter we look at what the injury leaves behind. That is the scar, which we call fibrosis, and the special stains that show it.

Every biopsy report you read as a fellow will end with a number for fibrosis. It may say "stage two", or "F three", or "Ishak four". The team will want to know what that number means for the child. Is the liver getting worse? Is there portal hypertension? Is this cirrhosis? And the report will often list special stains, with names such as trichrome, reticulin, rhodanine or PAS with diastase. Some of these stains are very useful. Others can quietly mislead you, especially in a young infant.

So this chapter asks two questions. First, how much fibrosis is there, and can I trust the number? Second, which stain answers my clinical question, and which stain can give a false answer in a child?

We will start with the shape of the scar. Then we will look at why a small biopsy gives the wrong answer. Then we will go through the scoring systems, and see why the same liver can get four different numbers. After that we will see that scars can go back. Then we will go through the stains, first those that show the scar, and then those that find a cause. We will finish with the traps of the infant liver, with what the guidelines say, and with four clinical cases.

A note on the sources. For liver biopsy in children, the main document is the position paper of the European paediatric society, ESPGHAN, written by Dezsőfi and colleagues in 2015. For fatty liver disease in children, we will use the American Association for the Study of Liver Diseases guidance of 2025 and the ESPGHAN paper of 2026. For Wilson disease, we will use the American guidance of 2022 and the European guideline of 2025. For the technical side of liver biopsy, we will use the British Society of Gastroenterology guideline of 2020.

Part one. The shape of the scar.

Fibrosis is laid down by two kinds of cells. The first are the stellate cells, which sit in the small space between the liver cells and the sinusoids. The second are the portal fibroblasts, which live in the portal tract. When the liver is injured, these cells become active and make collagen. They make it where the injury is. This gives us a simple and powerful idea. The shape of the scar tells you the route of the injury.

There are five main shapes to know.

The first is the hepatitic pattern. This is the scar of autoimmune hepatitis, hepatitis B and C, and Wilson disease. Inflammation starts in the portal tract, crosses the limiting plate at the interface, and moves into the lobule. The scar follows the same path. It starts in the portal tract, then spreads out, and then forms bridges from the portal tract to the central vein. Because the central veins are pulled into the scar early, blood flow through the liver is disturbed early. So in hepatitic disease, portal hypertension can appear before the liver is fully cirrhotic.

The second is the biliary pattern. This is the scar of biliary atresia, of MDR three deficiency caused by variants in ABCB4, of Alagille syndrome, of sclerosing cholangitis and of cystic fibrosis. Here the injury starts with swelling of the portal tract and a ductular reaction. The scar then links one portal tract to the next, forming bridges from portal tract to portal tract. The central veins are spared for a long time. The late liver has irregular nodules that fit together like the pieces of a jigsaw puzzle, and each nodule has a pale rim of swollen liver cells, which is the cholate stasis we met in the last chapter.

The third is the steatohepatitic pattern, the scar of fatty liver disease. In adults, the first scar is a fine net of collagen around individual liver cells in zone three, near the central vein. Pathologists call it perisinusoidal fibrosis, or chicken-wire fibrosis, because of how it looks. In children the pattern is often different. As we saw in the last chapter, Schwimmer and colleagues found in 2005 that fifty-one percent of children with fatty liver disease had the type two pattern, in which the injury sits around the portal tract. In these children, fibrosis often starts in the portal tract as well. One more point is important. Perisinusoidal fibrosis is almost invisible on the routine stain, called H and E. You need a collagen stain, such as a trichrome, to see it.

The fourth is the congestive pattern. This is the scar of the child after a Fontan operation, of constrictive heart physiology, and of Budd-Chiari syndrome. Here the blood cannot leave the liver easily. The sinusoids around the central vein become congested, and the scar starts there. It then forms bridges from one central vein to the next. The result looks like the normal liver turned inside out, and pathologists call it reverse lobulation. In 2014, Dai and colleagues showed that when the scar extends into the portal tracts as well, this marks a higher pressure in the right atrium.

The fifth is the developmental pattern. This is congenital hepatic fibrosis, often seen with autosomal recessive polycystic kidney disease. The bile ducts did not form properly before birth, which is called a ductal plate malformation, and the portal tracts are wide and scarred from the start. This is important. Congenital hepatic fibrosis is not cirrhosis. There are no regenerative nodules and liver function is normal. These children present with portal hypertension, often with

bleeding varices, but their liver can make proteins and clotting factors perfectly well.

So before you look at the number for fibrosis, look at the description of its shape. Portal to central suggests hepatitis. Portal to portal suggests bile duct disease. Central to central suggests congestion. And broad portal fibrosis without nodules in a child with kidney cysts suggests congenital hepatic fibrosis.

Part two. The smaller the core, the milder the disease.

A needle biopsy takes about one fifty-thousandth of the liver. So the first question about any fibrosis stage is whether the sample was big enough to trust.

The classic study on this question was published in 2003 by Colloredo and colleagues. They took one hundred and sixty-one biopsies from patients with chronic viral hepatitis, all at least three centimetres long. Then they shortened the same biopsies on the slide, first to one and a half centimetres, and then to one centimetre, and scored each length again.

The results were striking. With the full-length core, forty-nine point seven percent of biopsies were scored as mild for inflammation. When the same core was cut to one and a half centimetres, this rose to sixty point two percent. When it was cut to one centimetre, it rose to eighty-six point six percent. For fibrosis, the pattern was the same. Fifty-nine percent were scored as mild at full length, sixty-eight point three percent at one and a half centimetres, and eighty point one percent at one centimetre.

Notice what this means. The disease did not change. Only the length of the sample changed. A short core does not miss disease at random. It systematically makes the disease look milder than it is.

Other studies agree. In 2003, Bedossa and colleagues showed that the fibrosis stage was correct in only sixty-five percent of cores fifteen millimetres long, and in seventy-five percent of cores twenty-five millimetres long. In 2002, Regev and colleagues took biopsies from both the right and the left lobe of the same patients with hepatitis C. The two lobes differed by at least one stage in thirty-three point one percent of patients. And in 2005, Ratziu and colleagues found that two cores taken at the same time from patients with fatty liver disease differed in fibrosis stage in forty-one percent.

How long should a core be? The guidelines give different answers, and we will come back to this at the end. As a working rule, a core of about two centimetres with at least ten to eleven complete portal tracts is adequate. Fewer than eleven portal tracts tends to underestimate the stage. So when you read a report, look for

the length of the core and the number of portal tracts before you look at the stage. If they are not reported, ask for them.

Part three. The scoring systems.

There are many scoring systems for fibrosis. This is confusing, but it becomes easier once you know one fact. Every system was built for one adult disease. Children borrow them.

The Ishak system was published in 1995 for chronic viral hepatitis. It is also called the modified histological activity index. It stages fibrosis from zero to six, and grades inflammation from zero to eighteen. Its strength is that it has several fine steps for bridging fibrosis. Its weakness is that it was not designed for the portal-to-portal septa of biliary disease.

The METAVIR system was developed in France between 1994 and 1996 for hepatitis C. It stages fibrosis from F zero to F four, and grades activity from A zero to A three. Its weakness is at the top of the scale. F four includes every liver with cirrhosis, from early, thin-septum cirrhosis to end-stage disease.

The Batts-Ludwig system was published in 1995 for chronic hepatitis of any cause. It stages fibrosis from zero to four. Be careful here. Ludwig also published a staging system in 1978 for primary biliary cholangitis, and the words are similar, but it is a different system for a different disease.

The NASH Clinical Research Network system, published by Kleiner and colleagues in 2005, was built for fatty liver disease in adults and children. It stages fibrosis from zero to four, and it splits stage one into three parts. Stage one a is mild perisinusoidal fibrosis in zone three. Stage one b is moderate perisinusoidal fibrosis in zone three. And stage one c is fibrosis in the portal tract only. Stage one c is the paediatric pattern. The same system gives an activity score, called the NAFLD activity score, or NAS, from zero to eight.

The SAF score, with the FLIP algorithm, was developed in Europe between 2012 and 2014 for fatty liver disease in obese adults. SAF stands for steatosis, activity and fibrosis. Activity here means only two things: ballooning of the liver cells and inflammation inside the lobule.

The Nakanuma system, published in 2010, is for primary biliary cholangitis. It adds together three things: fibrosis, loss of bile ducts, and deposits of copper-binding protein seen on the orcein stain. So it includes cholestasis in the stage.

Finally, the Laennec system, published between 2009 and 2011, divides cirrhosis itself into three substages, called four A, four B and four C, according to how broad the septa are. It was developed in adults, mostly with alcohol-related disease, and has not been tested in children.

Now the important consequence. The same liver can receive four different numbers. Let us walk through it.

A liver with no fibrosis is stage zero in every system. A liver in which some portal tracts are expanded by fibrosis is Ishak one, METAVIR F one, Batts-Ludwig one and NASH Clinical Research Network one c. When most portal tracts are expanded, it becomes Ishak two, but it is still METAVIR F one. When there is occasional bridging from portal tract to portal tract, it is Ishak three, METAVIR F two, and stage three in both Batts-Ludwig and the NASH network system. When there is marked bridging, it is Ishak four, but METAVIR F three. Bridging with occasional nodules is Ishak five, and METAVIR F three to F four. And definite cirrhosis is Ishak six, METAVIR F four, and stage four in both of the other systems. Perisinusoidal fibrosis in zone three alone exists only in the NASH network system, as stage one a or one b.

Look at what this means for the word "four". Ishak four means marked bridging. METAVIR F four means cirrhosis. These are not the same liver. So there are two firm rules. Never average stages from different systems. And never compare two biopsies from the same child if they were scored in different systems. If you want to know whether the fibrosis has changed, ask for both biopsies to be scored in the same system, ideally by the same pathologist.

Part four. Cirrhosis is not one stage, and scars from different routes are not the same.

We usually think of cirrhosis as the end of the scale. But cirrhosis itself ranges from mild to very severe.

In 2011, Kim and colleagues studied one hundred and twenty-three patients with cirrhosis proven on biopsy, and measured the pressure in the portal system, using the hepatic venous pressure gradient. They used the Laennec system. In substage four A, with thin septa, the mean pressure gradient was eight point one millimetres of mercury. In four B, with at least two broad septa, it was twelve point four. In four C, with a very broad septum or very small nodules, it was sixteen point three. Remember that a gradient of ten or more is called clinically significant portal hypertension. So only the more advanced substages carried significant portal hypertension on average.

A second approach is to measure the scar with a computer. The Sirius red stain colours collagen, and software can calculate what percentage of the biopsy is collagen. This is called the collagen proportionate area. In the studies by Calvaruso and colleagues, the collagen proportionate area was thirteen percent in Ishak stage five and twenty-three percent in Ishak stage six. And in adults, only

the collagen proportionate area, not the Ishak stage, predicted a pressure gradient of ten or more. The best cut-off was fourteen percent.

The practical lesson is this. When a report simply says "F four" or "cirrhosis", ask the pathologist about the width of the septa, or ask for a collagen proportionate area if your laboratory can measure it. The portal pressure lives in the septa.

Now, what about fatty liver disease in children? The NASH network system was validated on fifty cases, and eighteen of them were children. It created stage one c for portal fibrosis. But the activity score, the NAS, does not include portal inflammation, which is prominent in the paediatric pattern. In 2012, Alkhouri and colleagues developed the Pediatric NAFLD Histological Score. It adds portal inflammation, scored from zero to two, with extra weight. It was built in two hundred and three Italian children and tested in one hundred more. It separated steatohepatitis from simple fatty liver very well, with an area under the curve of zero point nine six, and zero point nine four in the validation group. The European SAF score, in turn, helped non-expert pathologists agree with experts. Their agreement rose from forty-two percent to seventy-five percent when they used the algorithm.

The two newest guidelines, the American guidance of 2025 and the European paediatric paper of 2026, agree on the key points. They call the paediatric pattern zone one steatohepatitis. They ask the pathologist to report where in the lobule the changes are. And they both say that a score does not make the diagnosis. The activity score is a measure of activity, not a diagnosis of steatohepatitis. So for a child with fatty liver disease, ask for the NASH network stage, including one c, and one sentence on portal inflammation.

What about the biliary and congestive scars? Remember that the hepatitis scores assume the scar bridges from the portal tract to the central vein. Bile duct disease and heart failure scar along different lines.

For bile duct disease, the old Ludwig system of 1978 is still the default for primary biliary cholangitis and sclerosing cholangitis. It goes from portal, to periportal, to septal, to cirrhosis, and it looks only at fibrosis. The Nakanuma system adds duct loss and copper-binding protein, and sums them into a score from zero to nine, which is converted into a stage from one to four. But it is not easy to use. When twenty-eight pathologists staged the same slides, their agreement was only fair, with a kappa of zero point three eight five.

For biliary atresia, there is no specific score. METAVIR is the most used, and the median stage at the time of the Kasai operation is about F two. But the septa in atresia run from portal tract to portal tract and spare the veins. So a METAVIR F

three in biliary atresia and a METAVIR F three in hepatitis C do not carry the same risk of portal hypertension.

For the congested liver, there is a congestive hepatic fibrosis score. It moves from fibrosis around the central vein, to fibrosis that also involves the portal tract, to bridging. In a child after a Fontan operation, one more trap applies. Liver stiffness measured by elastography rises with the venous pressure itself, so it overestimates fibrosis in these patients.

Part five. Scars can go back.

For many years we thought of fibrosis as a one-way street. We now know that scars can recede.

In 2025, Pakarinen and colleagues reported the Finnish national cohort of children with biliary atresia. They studied one hundred and three pairs of biopsies, one taken at the Kasai operation and one taken later as a protocol biopsy. In forty percent of children, the fibrosis stage fell by one or two METAVIR stages after the Kasai. And this mattered. When the fibrosis decreased, the chance of keeping the native liver was much higher, with a hazard ratio of eleven point five. The confidence interval was wide, from one point five to eighty-seven, because the numbers were small, but the direction was clear.

In hepatitis B, a group in Beijing developed a way to describe whether the scar is growing or shrinking. It is called the P I R classification, which stands for predominantly progressive, indeterminate, and predominantly regressive. In 2017, Sun and colleagues studied seventy-one patients with paired biopsies before and after seventy-eight weeks of treatment with entecavir. Before treatment, thirteen percent of biopsies looked predominantly regressive. After treatment, seventy-eight percent did. And here is the key finding. Of the patients whose Ishak stage did not change, seventy-two percent were classed as regressive. The number stayed the same, but the biology had turned around.

What does a regressing scar look like? Wanless and colleagues described what they called the hepatic repair complex in 2000. It includes thin septa with holes in them, called perforated septa, isolated thick collagen fibres, remnants of portal tracts and hepatic veins, and very small nodules. When you see these words in a report, the scar is likely going back.

So the lesson is that the stage number can stand still while the scar is receding. If the question is whether a child is improving on treatment, ask the pathologist whether the septa look progressive or regressive.

Part six. Stains that show the scar.

The routine stain, H and E, shows injury very well. It shows inflammation, dead cells, fat and bile. But it shows scar poorly. For scar, the pathologist uses connective tissue stains. These stains show how much scar there is, and how old it is.

Masson trichrome stains collagen blue or green. It should be done on every biopsy that is being staged. It is essential for seeing the fine perisinusoidal fibrosis of steatohepatitis in zone three. The trap is that thick sections, and areas where the liver has collapsed, can make the fibrosis look worse than it is.

Sirius red stains collagen types one and three, and it can be measured by computer. This is the stain used for the collagen proportionate area and for dividing cirrhosis into substages. The trap here is that the error from sampling, which part of the liver the needle happened to hit, is larger than the error from reading.

The reticulin stain shows type three collagen, the fine frame that holds the liver cell plates. It is useful in three situations. First, it separates collapse from true fibrosis. When many liver cells die, as in acute liver failure, the framework falls together and can look like cirrhosis. This is called post-necrotic collapse, and it is the main trap. Second, reticulin helps to diagnose nodular regenerative hyperplasia, in which the liver becomes nodular without real fibrosis. Third, it helps to diagnose liver cancer, in which the plates become more than three cells thick and the reticulin frame is lost.

The orcein and Victoria blue stains show elastic fibres. Elastic fibres build up slowly. So an old septum is rich in elastic fibres, and a fresh area of collapse is not. These stains help to date a scar. They also stain two other things. One is the surface antigen of hepatitis B, and the other is copper-associated protein, which builds up in chronic cholestasis.

Finally, the cytokeratin stains CK seven and CK nineteen show bile ducts, ductules and so-called intermediate liver cells. They make duct counting much easier. When liver cells near the portal tract begin to stain for CK seven, this is a sign of chronic cholestasis. And one trap to remember from the last chapter: ductules are not ducts. Do not count the small ductules of a ductular reaction as true bile ducts.

Part seven. Stains that find a cause.

The second group of stains looks for a substance that points to a cause. There is a useful rule here. The simple chemical stains, called histochemistry, propose a

diagnosis. The antibody stains, called immunohistochemistry, and chemical measurements of the tissue, dispose of it, that is, they confirm or reject it.

The PAS stain shows glycogen. It is used when glycogen storage disease or glycogenic hepatopathy is suspected. The trap is that glycogen washes out of the tissue if it is fixed in a water-based fixative.

PAS with diastase is the same stain after the glycogen has been digested away. What remains pink are two things. The first are the globules of alpha one antitrypsin in deficiency of the Z type. The second are macrophages full of ceroid pigment, which are a marker of recent liver cell death. The trap is that globules are absent in young infants, as we will see in a moment. So the alpha one antitrypsin phenotype, measured in blood, makes the diagnosis, not the globules.

Rhodanine stains copper, and orcein stains copper-associated protein. These are used in chronic cholestasis and when Wilson disease is suspected. The trap is that they are negative in most cases of early Wilson disease, and that they are normally positive in infants under four months of age.

Perls stain shows iron, as haemosiderin, graded from zero to four. It is used for iron overload, and for gestational alloimmune liver disease, when it is also used on a biopsy of the inside of the cheek, called a buccal biopsy. The trap is that the newborn liver is naturally rich in iron. So iron in the liver of a newborn is not specific.

The antibody stains are more precise. They include stains for BSEP, the bile salt export pump, and for MDR three, both used in familial intrahepatic cholestasis. There is a stain for the Z form of alpha one antitrypsin. There is a stain for C five b nine, the membrane attack complex of complement, used in gestational alloimmune liver disease. And there is a stain for metallothionein, a copper-binding protein, used in Wilson disease. Each has its own trap. For example, C five b nine is positive in twenty-seven percent of livers from infants who do not have gestational alloimmune liver disease.

And finally, there is the chemical measurement of copper in the liver, reported in micrograms per gram of dry weight. It should be done on any biopsy where Wilson disease is possible. It needs a core that has not been put into formalin. And remember that any long-standing cholestasis also raises the liver copper.

Let us look more closely at copper, because this is where the stains fail most often. In 1998, Pilloni and colleagues studied seventy-nine biopsies from seventy-four patients with Wilson disease, aged four to sixty years. They used three copper stains: Timm silver, rhodanine and orcein. In the earliest stage, when the liver showed only fat, the stains were positive in only thirty point four percent,

thirteen point two percent and seventeen point five percent of biopsies. At the stage of interface hepatitis, they were positive in forty point one, twenty-six point seven and twenty-six point seven percent. Even at the stage of bridging fibrosis or cirrhosis, they were positive in only fifty-eight point six, thirty-six point six and twenty-nine point three percent.

Why is this? In early Wilson disease, copper is spread through the cytoplasm of the liver cells, where the stain cannot see it. Only later is it packed into lysosomes, where it becomes visible. So copper is highest early, but it stains worst early.

Measurement of copper is much better. In 2005, Ferenci and colleagues showed that a liver copper above two hundred and fifty micrograms per gram of dry weight had a sensitivity of eighty-three point three percent and a specificity of ninety-eight point six percent for Wilson disease. When the cut-off was lowered to seventy-five micrograms per gram, the sensitivity rose to ninety-six point five percent, and the specificity was still ninety-five point four percent.

So the rule is simple. A negative rhodanine stain does not lower the probability of Wilson disease at all. If Wilson disease is possible, send an unfixed core for copper measurement at the time of the biopsy.

Part eight. The infant liver.

The liver of a young infant is different. Normal infant physiology produces stains that look positive, and young disease produces stains that look negative. There are four traps to know.

The first trap concerns alpha one antitrypsin. In 1975, Talbot and Mowat studied infants with the ZZ type of deficiency. When the biopsy was taken before twelve weeks of age, none of them had the typical PAS with diastase globules. When it was taken after twelve weeks, all of them had. So in a cholestatic infant under twelve weeks, the absence of globules means nothing. The phenotype and the genes make the diagnosis.

The second trap concerns copper and iron. The ESPGHAN paper of 2015 reminds us that stainable copper-associated protein and iron are normal in the infant liver until about four months of age. Blood-forming cells in the liver, called extramedullary haematopoiesis, are normal until about six months. So copper, iron and blood-forming cells must be read against the age of the child.

The third trap concerns the BSEP stain. It is a very useful stain, but it is not perfect. Seventy-five to ninety percent of livers with BSEP deficiency are negative on the stain. So a normal BSEP stain does not exclude the disease, especially when the gene variant changes only one amino acid, which is called a missense

variant. In that case the protein may still be made and still stain, but it does not work.

The fourth trap concerns gestational alloimmune liver disease. In 2017, Dubruc and colleagues studied the C five b nine stain. It was positive in every liver with this disease, but in only twenty-six percent did it cover more than seventy-five percent of the liver tissue. And it was also positive in twenty-seven percent of livers without this disease, including viral hepatitis, a mitochondrial disease caused by variants in DGUOK, and bile acid synthesis defects. So C five b nine is sensitive, but not specific. The diagnosis still rests on the whole picture, including iron outside the liver, for example on the buccal biopsy or on magnetic resonance imaging.

The lesson for the cholestatic infant is to send the alpha one antitrypsin phenotype and the gene panel, and to read copper, iron and globules against the age of the baby.

Part nine. What the guidelines say.

How do the guidelines compare on fibrosis and stains? There are seven points worth knowing.

First, the size of the core. Here the guidelines differ. The ESPGHAN paper asks for at least two centimetres or at least ten portal tracts in suspected biliary atresia, and more than two centimetres in fatty liver disease. The American guidance on liver biopsy asks for two to three centimetres with a sixteen gauge needle, and at least eleven complete portal tracts. The European and British documents ask for at least twenty millimetres with a sixteen gauge needle, and the Royal College of Pathologists sets a minimum of ten millimetres and six portal tracts.

Second, which staging system to use. Here there is no data to guide us. None of the documents names a single system. The British guideline asks for a recognised system suited to the disease, according to local practice. The American fatty liver guidance says only that scores cannot diagnose steatohepatitis on their own.

Third, the paediatric pattern of fatty liver disease. Here the guidelines agree. Both the American and the European paediatric documents call it zone one steatohepatitis and ask the pathologist to report where the changes are.

Fourth, the copper threshold for Wilson disease. Here the guidelines also agree. A liver copper above two hundred and fifty micrograms per gram of dry weight is diagnostic, or highly suggestive. And all of them remind us that cholestasis also raises the copper.

Fifth, how to handle the copper core. Here the European documents add something. They ask for a separate core, not put into formalin or saline, placed on moist filter paper or in a copper-free container. The American paper of 2009 on liver biopsy does not address this.

Sixth, the normal values in infants. Only the paediatric document addresses them. Copper-associated protein and iron are normal until four months, and blood-forming cells until six months.

Seventh, who should read the biopsy. Here the British guideline adds something. It asks for a pathologist experienced in liver disease, and it cites a review in which a second reading by an expert found a major discrepancy in twenty-eight percent of biopsies. The ESPGHAN paper says that a biopsy for biliary atresia should be read by a paediatric pathologist.

Part ten. How an expert reads the report.

Let us bring this together into four habits.

The first habit is to read the length first. A biopsy with fewer than eleven portal tracts under-stages the disease. A one-centimetre core calls about eighty percent of livers mild. So always look for the length and the number of portal tracts, and ask for them if they are missing.

The second habit is to remember that scores are only ordered categories. Ishak four is not METAVIR F four. F four hides the whole range from four A to four C. And the NAFLD activity score measures activity. It does not diagnose steatohepatitis.

The third habit is to use the stains to date the scar. Elastic fibres mean an old septum. Reticulin separates collapse from cirrhosis. And liver cells that stain for CK seven mean chronic cholestasis.

The fourth habit is to remember that a negative stain is not a normal result. Rhodanine misses early Wilson disease. PAS with diastase is blank before twelve weeks of age. And copper and iron are normal in the infant liver until four months.

Put simply, read the stage together with the length of the core, the name of the system and the age of the child, all in the same sentence.

To finish the teaching part of the chapter, here are the common clinical questions, and the stain or score that answers each one.

If you want to know how much fibrosis there is, ask for a trichrome or Sirius red stain and a score suited to the disease, on a core of at least two centimetres with ten to eleven portal tracts.

If you want to know how severe the cirrhosis is, ask for the Laennec substage or the collagen proportionate area. In adults, a collagen area of fourteen percent or more suggests clinically significant portal hypertension. And ask for reticulin and elastic stains to make sure the "cirrhosis" is not simply collapse.

If you want to know whether the fibrosis is regressing, ask whether the septa look progressive or regressive, and look for the features of the repair complex.

If you want to know whether a child has steatohepatitis, ask for the NASH network stage including one c, and a comment on portal inflammation. Remember that an activity score of five or more is not a diagnosis.

If you want to know whether the child has Wilson disease, send an unfixed core for copper measurement.

If you want to know whether an infant has alpha one antitrypsin deficiency, the globules will help only after twelve weeks. The phenotype decides.

If you suspect gestational alloimmune liver disease, look for iron outside the liver, and remember that C five b nine is sensitive but not specific.

And if you suspect a genetic cholestasis, ask for the BSEP, MDR three and CK seven stains, and electron microscopy, but remember that normal staining does not exclude a missense variant. Send the gene panel.

Part eleven. Four clinical cases.

Case one. A thirteen-year-old boy with obesity has had a raised ALT for two years. His biopsy is nine millimetres long and contains five portal tracts. The report says: fat in sixty percent of liver cells, mild portal inflammation, NAS three, fibrosis stage one c. The conclusion is "fatty liver, not steatohepatitis, minimal fibrosis". What do you think? There are two problems. First, the core is short. With only five portal tracts, the stage is likely to be too low, perhaps by a whole stage. Second, the conclusion is based on the activity score, which ignores portal inflammation, and the paediatric type two pattern is portal. Ask the pathologist whether the changes are zone one, and ask for a comment on the portal inflammation. Treat this boy as having possible steatohepatitis with possibly more fibrosis than reported, and use the biopsy together with the clinical picture, not the score alone.

Case two. A fifteen-year-old girl has an ALT of three hundred, a slightly low ceruloplasmin and interface hepatitis on the biopsy. The rhodanine stain is negative, and the report says "no evidence of copper overload". The team is ready to diagnose autoimmune hepatitis and start steroids. What do you advise? A negative rhodanine does not lower the chance of Wilson disease. In the Pilloni series, rhodanine was negative in about three quarters of biopsies at the stage of

interface hepatitis. Ask whether an unfixed core was kept for copper measurement. If not, send a twenty-four-hour urine copper, an eye examination for Kayser-Fleischer rings and ATP seven B gene testing. A liver copper above two hundred and fifty micrograms per gram of dry weight would make the diagnosis. Next time, always take the copper core at the time of the biopsy.

Case three. A seven-week-old girl has conjugated jaundice. Her biopsy shows giant cells, a ductular reaction and some iron in the liver cells. The PAS with diastase stain shows no globules, and the report says "alpha one antitrypsin deficiency not supported". The copper-associated protein stain is mildly positive. What next? At seven weeks, the absence of globules means nothing. In the Talbot and Mowat series, no infant with the ZZ type had globules before twelve weeks. Send the alpha one antitrypsin level and phenotype. The iron and the copper-associated protein are also normal at this age and should not be read as disease. This baby needs the full cholestasis work-up, including the phenotype and a gene panel.

Case four. A sixteen-year-old boy had a Fontan operation at the age of four. His liver stiffness on elastography is very high. His biopsy shows fibrosis around the central veins, with bridges from central vein to central vein, and some portal fibrosis. The report scores it as Ishak five, with the comment "approaching cirrhosis". The cardiology team asks whether he now has cirrhosis. What do you say? This is the congestive pattern, not the hepatitic one. The Ishak system assumes bridging from the portal tract to the central vein and was not built for this scar. Ask the pathologist to describe the pattern and to use a congestive hepatic fibrosis score. The portal fibrosis is important, because it marks a higher right atrial pressure, and this should go back to the cardiologists. And remember that liver stiffness overestimates fibrosis in Fontan circulation, because the congestion itself makes the liver stiff.

Key points to remember.

The shape of the scar names the route of injury. Portal to central is hepatitic. Portal to portal is biliary. Central to central is congestive. Broad portal fibrosis without nodules is congenital hepatic fibrosis, which is not cirrhosis.

In children with fatty liver disease, fibrosis often starts in the portal tract. Perisinusoidal fibrosis cannot be seen without a trichrome stain.

A short core under-stages. Cut from three centimetres to one, the share of biopsies scored as mild rose from about sixty to eighty percent. Ask for at least two centimetres and ten to eleven portal tracts.

Every score was built for one adult disease. Ishak four is marked bridging, but METAVIR F four is cirrhosis. Never compare biopsies scored in different systems.

Cirrhosis is not one stage. Broad septa and a collagen proportionate area of fourteen percent or more point to clinically significant portal hypertension.

For a child with fatty liver disease, ask for the NASH network stage including one c, and a comment on portal inflammation. Scores do not make the diagnosis.

A METAVIR F three in biliary atresia and in hepatitis C do not carry the same risk. In a child after a Fontan operation, elastography overestimates fibrosis.

Scars can recede. After the Kasai operation, forty percent of children had less fibrosis on a later biopsy, and they were more likely to keep their own liver. The stage can stay the same while the scar is regressing.

Trichrome and Sirius red show collagen. Reticulin separates collapse from cirrhosis. Elastic stains date a scar. CK seven shows ducts and chronic cholestasis.

A negative rhodanine does not exclude Wilson disease. Send an unfixed core. A liver copper above two hundred and fifty micrograms per gram of dry weight is diagnostic.

In infants, globules of alpha one antitrypsin are absent before twelve weeks, and copper and iron are normal until four months. The phenotype and the genes decide.

C five b nine is sensitive but not specific for gestational alloimmune liver disease. A normal BSEP stain does not exclude BSEP deficiency.

Read the stage together with the length of the core, the system used and the age of the child.

Questions to test yourself.

Question one. A biopsy shows bridges from portal tract to portal tract, with central veins spared and nodules with a pale rim. Which route of injury does this suggest? Answer: the biliary route, as in biliary atresia, sclerosing cholangitis, MDR three deficiency, Alagille syndrome or cystic fibrosis.

Question two. Why can congenital hepatic fibrosis cause variceal bleeding in a child with normal liver function? Answer: because the broad portal fibrosis causes portal hypertension, but this is not cirrhosis. There are no regenerative nodules, and the liver cells work normally.

Question three. What happened to the fibrosis stage when Colloredo and colleagues shortened the same biopsy from three centimetres to one? Answer: the share of biopsies scored as mild rose from fifty-nine percent to about eighty percent. A short core systematically under-stages.

Question four. One biopsy is reported as Ishak four and a later one as METAVIR F three. Has the fibrosis improved? Answer: you cannot tell. Ishak four and METAVIR F three describe roughly the same degree of marked bridging. The two biopsies must be scored in the same system to compare them.

Question five. What is stage one c in the NASH Clinical Research Network system, and why does it matter in children? Answer: it is fibrosis in the portal tract only. It matches the type two, portal pattern of fatty liver disease, which is common in children.

Question six. A child's fibrosis stage has not changed on a repeat biopsy after treatment. Does that mean the disease has not improved? Answer: no. In the Beijing study of hepatitis B, seventy-two percent of patients with an unchanged Ishak stage had a scar that looked regressive. Ask whether the septa look progressive or regressive.

Question seven. Why are copper stains so often negative in early Wilson disease? Answer: because early copper is spread through the cytoplasm, where the stain cannot see it. It becomes stainable only later, when it is packed into lysosomes. Quantitative copper on an unfixed core is the test.

Question eight. A six-week-old infant with cholestasis has no PAS with diastase globules on the biopsy. Does this exclude alpha one antitrypsin deficiency? Answer: no. Globules are absent before twelve weeks of age, even in infants with the ZZ type. Send the phenotype.

End of chapter.